William Kwako

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EDUCATION

M.S., Computer Science – Interactive Intelligence, Georgia Tech, GPA: 4.0 August 2023 – August 2026
B.A., Physics, Willamette University May 2019

PROFESSIONAL EXPERIENCE

Technical Solutions Engineer – Epic Systems Corporation December 2019 – December 2022

- Served as point of contact for all claims-related issues across five health system clients; led weekly and biweekly meetings with revenue cycle analysts and managers to triage and resolve issues
- Deployed, maintained, and debugged revenue cycle software, including HIPAA-mandated EDI claims and remittance transactions (837I/837P, 835)
- Coordinated large-scale claims implementations (claim attachments, PLBs, rapid reconciliation, state reporting) to meet standardized claim conformance requirements
- Wrote M/MUMPS queries against production healthcare databases containing PHI to diagnose client issues and analyze data for resolution
- Headed the Debugging Task Force; trained new hires on internal debugging tools and revised the onboarding curriculum

Teaching Assistant – Willamette University, Physics Department January 2019 – May 2019

- Guided students through computational and experimental Introductory Physics 2 laboratory exercises
- Answered questions about class material and evaluated students' performance

Intern – Willamette University, Physics Department May 2018 – July 2018

- Developed the initial wavelet-based signal-detection pipeline (FFT convolution in Python), the first application of this method to the 2MASS Redshift Survey; later completed as my senior thesis
- Built a computer for parallel GPU processing to support analysis of a large-scale cosmological dataset

TECHNICAL SKILLS

- **Languages**: Python, C#, Java, HTML, CSS, JavaScript, SQL, M/MUMPS
- **AI & ML:** NumPy, Matplotlib, Stable-Baselines3, Gymnasium, PyTorch, scikit-learn, OpenAI API, Anthropic API, Prompt Engineering, LangGraph, Chroma, RAG
- **Frameworks & Tools:** Django, Git, Unity, LaTeX, .NET, Android Studio, dnSpy
- **Healthcare:** Epic, revenue cycle, X12 EDI (837I/837P, 835), HIPAA/PHI, claims and remittance workflows

PROJECTS AND RESEARCH

- **Benchmarking Reinforcement and Imitation Learning on a Deterministic Platformer: Jump King**
  Reinforcement Learning Research, 2026

- Built a full pipeline interfacing with a live commercial game through real-time state extraction (custom C# mod), TCP socket communication, and a custom Gymnasium environment
- Designed and implemented behavioral cloning (BC) from human playthroughs, per-level reward shaping, and a class-based framework for training and evaluating RL, BC, and BC+RL agents
- Chained per-level agents reliably complete the full 43-level base game in ~10 minutes
- Showed BC initialization solves levels where pure RL fails entirely, and converges far faster elsewhere
- **Tools**: Python, C#, Stable-Baselines3, Gymnasium, dnSpy, .NET

- **Predicting Speedrunner Retention from Activity Patterns**
  Applied Machine Learning Research, 2026
  - Built an end-to-end pipeline from the speedrun.com API into SQLite, removing unusable records and engineering behavioral features (recency, frequency, run density, consistency)
  - Compared three model families (logistic regression, gradient boosting, and a PyTorch MLP) across three games, reaching ~0.79 AUC on splits controlled for temporal and identity leakage
  - Showed standard grouped cross-validation inflated AUC by ~0.09 (0.88 vs. 0.79), largely reflecting temporal leakage
  - Used Spearman correlations to show a predictive feature's near-zero importance was a collinearity artifact, not a lack of signal
  - **Tools**: Python, PyTorch, scikit-learn, NumPy, SQLite

- **Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation**
  Cosmology Research – Senior Thesis, Willamette University, 2018-2019
  - Built a signal-detection pipeline applying 3D wavelet transforms via FFT convolution to a noisy cosmological density field (2MASS Redshift Survey)
  - Implemented preprocessing (Gaussian smoothing, zero-padding) and a custom peak-detection algorithm to isolate candidate signals
  - Validated detections against a random-sampling null distribution, identifying three signals at 4.31σ (0.0008% probability by chance)
  - **Tools**: Python, NumPy, SciPy, Matplotlib

- **Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems**
  Educational Technology Research, 2025-2026
  - Built a web application generating personalized coding exercises to improve code comprehension
  - Designed and implemented a multi-model pipeline with conditional routing, prompt chaining, guardrails, and feedback loops across OpenAI and Anthropic APIs
  - Built a validation and repair loop that executes generated code, verifies it against user specifications via AST inspection, and escalates to a stronger model on failure
  - **Tools**: Python, Django, JavaScript, HTML/CSS, OpenAI and Anthropic APIs

- **Improving Users' Academic Writing Ability Through LLM-Generated Feedback**
  Educational Technology Research, 2026
  - Built a Django web application that generates writing feedback by decomposing analysis across six specialized LLM agents using LangGraph, with a synthesis step reconciling critiques
  - Designed a controlled evaluation (position-bias and cross-family-judge controls) that isolated analyzer strength as the performance bottleneck

- Implemented RAG with Chroma (genre exemplars, rubric criteria) and tested it via ablation; found no net benefit at n=10 and reported it as a null result
- **Tools**: Python, Django, LangGraph, Chroma, HTML/CSS, OpenAI and Anthropic APIs

POSTERS AND PRESENTATIONS

- *Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems*
  Poster Presentation – National Council on Measurement in Education (April 2026)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*
  Oral Presentation – Willamette-Linfield intercollegiate symposium (May 2019)
  Poster Presentations – 223rd Meeting of the American Astronomical Society (January 2019), Murdock College Science Research Conference (November 2018)